

INTRODUCTION TO ALGORITHMS

3-2-0-4

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Reference: **INTRODUCTION TO ALGORITHMS**, by Cormen, Leiserson, Rivest & Stein, PHI. [//mitpress.mit.edu/algorithms](http://mitpress.mit.edu/algorithms)

We shall refer to this book as [CLRS].

Stress will be on the practical aspects of algorithms.

Based on [CLRS]:

... an algorithm is a *well-defined computational procedure* that takes as **input** a *set of values* and produces as **output** another specified *set of values* ...

... a sequence of computational steps that transforms as specified the **input** into **output**.

Clearly, we need a notation to define – that is, specify – an algorithm. We shall use the notation of **pseudocode** for this purpose.

Suppose we need to produce in output the **sorted** set of values in input.

Question: When is it possible to sort a set of values?

Hint: Can we compare apples and oranges? An ordering relation must exist.

Let us consider a simple sorting algorithm.

```

INSERTION-SORT (A)
for j = 2 to length[A]
    key = A[j]
    // Insert A[j] into sorted sequence A[1..j-1]
    i = j-1
    while i > 0 and A[i] > key
        A[i+1] = A[i]
        i = i-1
    A[i+1] = key

```

NOTE:

1. The notation used is different from that used in [CLRS]. The above notation pseudocode does not include **do**, curly brackets or semi-colons.
2. Red font → elements are being compared.
3. Importance of indentation. Note the differences.
4. This sort algorithm works in-place.
5. Learn the importance of testing for end values of loop variable.

Loop invariant: Every time we enter the **for** loop, the sequence **A[1..j-1]** is sorted.

IMPORTANT: Get used to analysing algorithms “on paper” – that is, even when you are not programming the algorithm.

QUESTION 1: How much time will this algorithm take for sorting 10, 100, 1000 ... 10^6 elements?

QUESTION 2: Do we have enough information to answer the above question?

Think about such questions related to algorithms.